

Q3 Constant acceleration equations

Name: _____

Class: _____

Date: _____

Time: **460 minutes**

Marks: **376 marks**

Comments:

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Q1.

A rabbit runs in a horizontal straight line ABC across a field.

- (a) The rabbit runs from rest at A with a constant acceleration of 0.8 m s^{-2} and reaches B after 3 seconds.

Find its speed at B .

(2)

- (b) The rabbit then runs from B with constant speed and reaches C after a further 4 seconds.

Sketch a velocity-time graph of the motion of the rabbit as it runs from A to C .

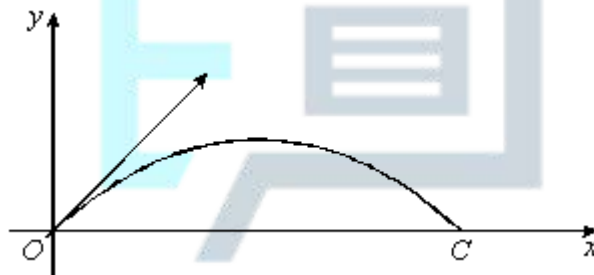
(3)

- (c) Find the average speed of the rabbit as it runs from A to C .

(5)

(Total 10 marks)

Q2.



Oliver throws a ball from point O , with velocity $\begin{bmatrix} 7 \\ 7 \end{bmatrix} \text{ m s}^{-1}$. The ball is subsequently caught by

Chris at a point C which is on the same horizontal level as O , as shown in the diagram. During the motion from O to C the ball moves in a vertical plane.

- (a) Taking O as the origin, find the position vector of the ball t seconds after being thrown.

(3)

- (b) Show that the equation of the path of the ball is

$$y = x - 0.1x^2,$$

where (x, y) are the coordinates at time t of the ball relative to the origin O .

(4)

- (c) A horizontal wire, which is part of a fence, cuts the plane of motion of the ball at the

point (4, 2).

Determine whether the ball passes above or below the wire.

(3)

(d) Find the distance OC .

(4)

(e) Find the greatest speed of the ball during its motion from O to C .

(2)

(Total 16 marks)

Q3.

A holiday-maker drops a book from a balcony of her hotel room. The book falls vertically until it hits the ground.

(a) The book falls from a height of 10 metres above the ground.

Find the speed with which it hits the ground.

(2)

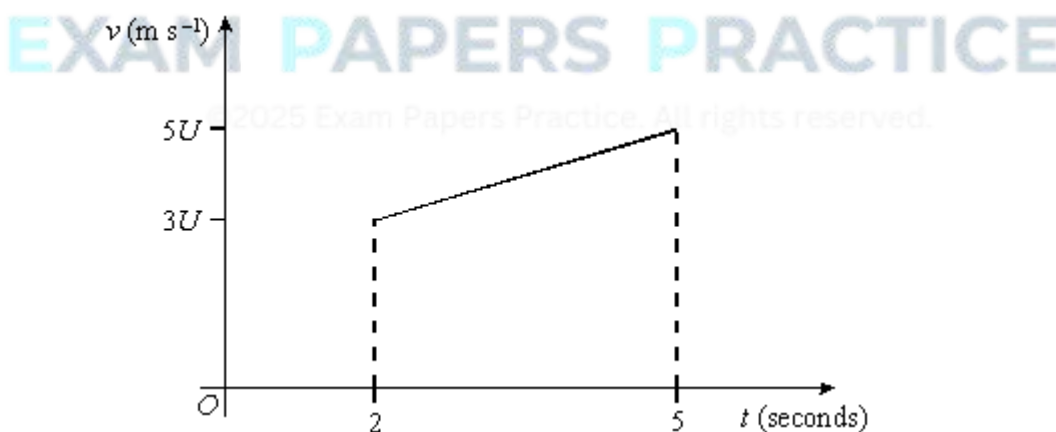
(b) State **one** modelling assumption you have made regarding the motion of the book as it falls.

(1)

(Total 3 marks)

Q4.

The velocity–time graph shows the motion of a particle P moving with constant acceleration.



At times $t = 2$ and $t = 5$, P has velocities $3U$ and $5U$ respectively.

(a) (i) Find, in terms of U , the acceleration of P .

(2)

(ii) Find, in terms of U , the distance travelled by P between the times $t = 2$ and $t = 5$.

(2)

- (b) When $t = 5$, the motion of P changes and subsequently P moves with a constant retardation. The particle P comes to rest after travelling a **further** 20 metres in the next 4 seconds.

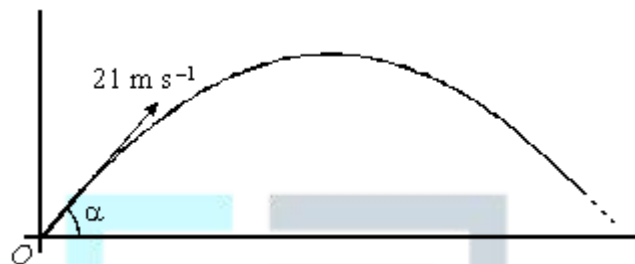
Find the value of U .

(3)

(Total 7 marks)

Q5.

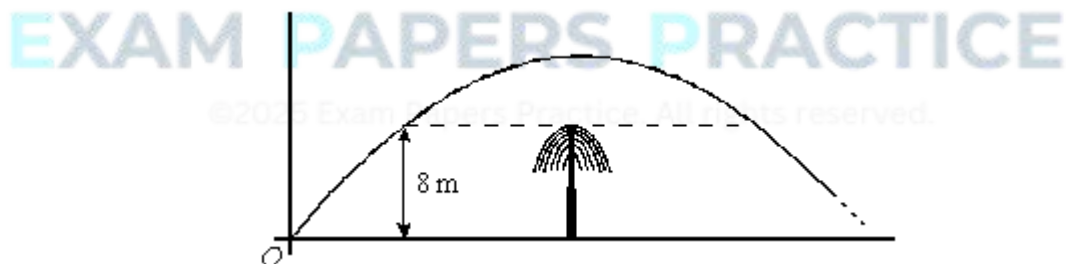
Paul throws a ball from point O with a velocity 21 m s^{-1} at an angle of α to the horizontal, where $\sin \alpha = 0.7$. The ball subsequently moves freely under gravity in a vertical plane, as shown in the diagram.



- (a) Show that the time taken for the ball to reach its greatest height above O is 1.5 seconds.

(3)

- (b) When the ball reaches its greatest height, it passes over a tree of vertical height 8 metres, as shown in the diagram below.



- (i) Find the vertical distance between the ball and the top of the tree at this time.

(4)

- (ii) Find the time between the ball leaving O and first reaching the horizontal level of the top of the tree. Give your answer to two decimal places.

(5)

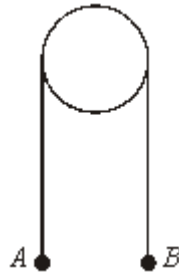
- (iii) Find the length of time for which the ball is above the horizontal level of the top of the tree.

(2)

(Total 14 marks)

Q6.

Two particles, A and B , are connected by a light inextensible string which passes over a smooth fixed peg, as shown in the diagram.



The particles A and B have masses 0.21 kg and 0.14 kg respectively. The system is released from rest. It may be assumed that, during the motion, neither particle reaches the peg.

- (a) Show that the magnitude of the acceleration of the particles during the subsequent motion is 1.96 m s^{-2} . (5)
- (b) Find the speed of the particles after 2 seconds of the motion. (2)
- (c) After 2 seconds of the motion, the string breaks and the particles move independently under the influence of gravity alone. Find the time between the string breaking and the particle B coming to instantaneous rest. (2)
- (d) Sketch a velocity-time graph to show the motion of B between being released from rest and subsequently coming instantaneously to rest. (3)

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(Total 12 marks)

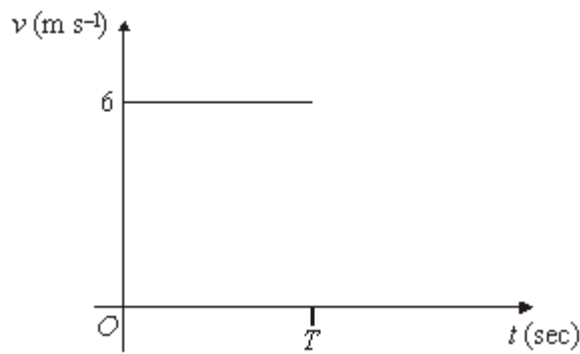
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Q7.

Fiona is cycling along a straight horizontal road when she spots her friend, Marie, ahead of her. Marie is jogging along the same road and in the same direction.

Fiona is cycling with constant speed 6 m s^{-1} and Marie is jogging with constant speed 3 m s^{-1} .

- (a) The velocity-time graph below shows Fiona's motion from the moment when she spots Marie, when $t = 0$, until she reaches a bridge, when $t = T$.



Marie takes 2 seconds longer than Fiona to reach the bridge.

Copy the graph and draw another line to show Marie's motion from $t = 0$ until she reaches the bridge. Label the axes appropriately.

(2)

(b) When $t = 0$, Marie is x metres from the bridge and Fiona is 20 metres behind her.

(i) Write down, in terms of x , the distance travelled by Fiona.

(1)

(ii) Write down an equation in terms of x and T for Fiona's motion.

(1)

(iii) Write down an equation in terms of x and T for Marie's motion.

(2)

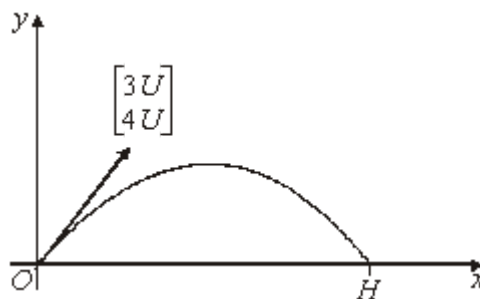
(iv) Hence find the values of x and T .

(4)

(Total 10 marks)

Q8.

George throws a ball from a point O , with velocity $\begin{bmatrix} 3U \\ 4U \end{bmatrix}$. The ball subsequently lands at a point H , which is at the same horizontal level as O , as shown in the diagram.



(a) Show that the time taken by the ball to travel from O to H is $\frac{8U}{g}$.

(3)

- (b) Find, in terms of g and U , the distance OH .

(2)

- (c) Find, in terms of U , the initial speed of the ball.

(2)

- (d) Find, in terms of g and U , the two times during the flight from O to H when the ball is moving with speed $\sqrt{18}U$.

(6)

(Total 13 marks)

Q9.

A van accelerates uniformly from rest on a straight horizontal road. After it has travelled 80 metres, its speed is 20 m s^{-1} .

- (a) (i) Find the time taken for the van to travel the 80 metres.

(2)

- (ii) Show that the acceleration of the van is 2.5 m s^{-2} .

(2)

- (b) The mass of the van is 1200 kg. The force produced by its engine has magnitude F newtons and acts on the van in its direction of motion.

- (i) Find F , if there is no resistance to motion.

(2)

- (ii) Find F , if there is a constant resistance force of 400 N.

(2)

(Total 8 marks)

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Q10.

A car is travelling along a straight horizontal road at a speed of 18 m s^{-1} when the driver sees a red traffic light ahead. The car then slows down uniformly, so that 6 seconds later the speed of the car is 9 m s^{-1} .



- (a) While the car is slowing down:

- (i) show that its acceleration is -1.5 m s^{-2} ;

(3)

- (ii) find the distance that it travels. (2)
- (b) After the car has slowed down for 6 seconds, the traffic lights change to green. The car then accelerates at a constant 1.2 m s^{-2} and passes the traffic lights at a speed of 15 m s^{-1} .
- (i) Show that the car accelerated for 5 seconds before reaching the traffic lights. (2)
- (ii) Find the distance of the car from the traffic lights when the car started to accelerate. (3)
- (c) Find the distance of the car from the traffic lights when the car began to slow down. (1)
- (Total 11 marks)

Q11.

A particle moves with constant acceleration in a horizontal plane. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the plane. At time $t = 0$, the particle is at rest at the origin. At time $t = 4$ seconds, the position vector of the particle is $(-16\mathbf{i} + 16\mathbf{j})$ metres.

- (a) Show that the acceleration of the particle is $(-2\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. (4)
- (b) Find the speed of the particle when $t = 5$. (4)
- (c) The mass of the particle is 3 kg. Two horizontal forces $\mathbf{P}$ and $\mathbf{Q}$ act on the particle. If $\mathbf{P} = 20\mathbf{i} - 10\mathbf{j}$, find $\mathbf{Q}$. (4)
- (Total 12 marks)

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Q12.

During a football match, a player kicks a ball from rest towards a goal, which is 12 metres from the ball. The ball is kicked with an initial velocity of 25 m s^{-1} at an angle of 17° above the horizontal.

- (a) Show that the time that it takes the ball to travel the horizontal distance of 12 metres is 0.502 seconds, correct to three significant figures. (3)
- (b) The ball hits the crossbar of the goal. Calculate the height of the crossbar. (3)
- (c) Find the angle between the velocity of the ball and the **vertical** just before the ball hits the crossbar. (5)

- (d) State two modelling assumptions that you have made.

(2)

(Total 13 marks)

Q13.

A car is travelling along a straight horizontal road. When the car is at a distance of 48 metres from a set of traffic lights, it is travelling at 20 m s^{-1} . The car slows down uniformly so that it has a speed of 4 m s^{-1} as it passes the traffic lights.

- (a) Show that the acceleration of the car is -4 m s^{-2} .

(2)

- (b) How far past the traffic lights does the car travel before it stops?

(3)

- (c) Calculate the total time that it takes the car to stop.

(2)

- (d) The car has mass 1100 kg. As the car slows down, it experiences a resistance force that has a constant magnitude of 200 newtons, and a force due to the brakes of magnitude F newtons.

Find F .

(3)

(Total 10 marks)

Q14.

A particle moves with constant acceleration $(4\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. At time $t = 0$, the particle is at the origin and has velocity $(3\mathbf{i} - 10\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Show that the velocity of the particle at time t seconds is

$$(3 + 4t)\mathbf{i} + (2t - 10)\mathbf{j}$$

(2)

- (b) Find the time when the particle is travelling due east.

(2)

- (c) (i) Find the position vector of the particle when $t = 10$.

(3)

- (ii) Find the velocity of the particle when $t = 10$.

(1)

- (d) The particle stops accelerating when $t = 10$ and then moves with constant velocity.

Find the distance of the particle from the origin after it has been travelling with this constant velocity for a further 10 seconds.

(6)

(Total 14 marks)

Q15.

A ball is projected vertically upwards from a height of 5 metres above ground level. Its initial speed is 7 m s^{-1} . Assume that the ball is a particle and that no resistance forces act on it as it moves.

- (a) Show that the maximum height of the ball is 7.5 metres above ground level. (3)
- (b) Show that the time that it takes the ball to reach its maximum height is 0.714 seconds, correct to three significant figures. (2)
- (c) The ball is caught when it is at a height of 2 metres above ground level.
 - (i) Find the total time for which the ball is moving. (4)
 - (ii) Find the speed of the ball when it is caught. (3)

(Total 12 marks)

Q16.

The diagram shows two particles, A and B , which are connected by a light inextensible string. The string passes over a smooth light pulley. Particle A has mass 20 kg and rests on a rough horizontal surface. The coefficient of friction between the particle and the surface is 0.15. Particle B has mass 5 kg and hangs with the string vertical, as shown in the diagram.



The particles are released from rest and begin to move.

- (a) Find the magnitude of the friction force acting on A . (2)
- (b) By forming an equation of motion for each particle, show that the acceleration of each particle has magnitude 0.784 m s^{-2} . (5)
- (c) Find the tension in the string. (2)

- (d) Find the distance that A moves in the first 3 seconds of its motion. Assume that the acceleration of A remains constant during this time.

(3)

(Total 12 marks)

Q17.

A ball is kicked from ground level so that its initial velocity is 10 m s^{-1} , at an angle of 60° above the horizontal. It hits a wall, which is at a distance of 8 metres from the initial position of the ball.

- (a) Show that the ball hits the wall 1.6 seconds after it was kicked.

(3)

- (b) Find the height of the ball as it hits the wall.

(3)

- (c) Find the speed of the ball as it hits the wall.

(5)

(Total 11 marks)

Q18.

A particle is initially at the origin and has velocity $(4\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$. It moves with constant acceleration, and 10 seconds later has velocity $(5\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Show that the acceleration of the particle is $(0.1\mathbf{i} - 0.5\mathbf{j}) \text{ m s}^{-2}$.

(3)

- (b) Find an expression for the position vector of the particle at time t seconds.

(2)

- (c) Find the time when the particle is due east of the origin.

(4)

- (d) Find the time when the particle is moving due east.

(4)

(Total 13 marks)

Q19.

A cyclist travels along a straight horizontal road. She accelerates uniformly from a speed of 2 m s^{-1} to 6 m s^{-1} as she travels 10 metres.

- (a) (i) Show that the acceleration of the cyclist is 1.6 m s^{-2} .

(2)

- (ii) Find the time that it takes the cyclist to travel this distance.

(2)

- (b) Model the cyclist as a particle of mass 65 kg. A constant resistance force of magnitude 35 N acts on the cyclist. A horizontal force of magnitude F newtons also acts on the cyclist in the direction of motion. Find F .

(3)

(Total 7 marks)

Q20.

At time $t = 0$, a boat has velocity $(2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-1}$ and is at the origin. After moving with a constant acceleration of $\mathbf{a} \text{ m s}^{-2}$ for 20 seconds, its position vector is $(60\mathbf{i} + 20\mathbf{j})$ metres. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find $\mathbf{a}$.

(3)

- (b) Find the velocity of the boat at time t seconds.

(2)

- (c) Find the speed of the boat when it is travelling south east.

(5)

(Total 10 marks)

Q21.

A car travels along a straight horizontal road. When it passes a set of traffic lights it is travelling at a speed of 3 m s^{-1} . The car then accelerates at a constant 1.2 m s^{-2} until it reaches a speed of 9 m s^{-1} .

- (a) Show that the car accelerates for 5 seconds.

(2)

- (b) Find the distance that the car travels as it accelerates from 3 m s^{-1} to 9 m s^{-1} .

(2)

- (c) The mass of the car is 1200 kg. Calculate the magnitude of the resultant force acting on the car while it is accelerating.

(2)

(Total 6 marks)

Q22.

A rough plane is inclined at an angle of 40° to the horizontal. A particle, of mass 5 kg, is sliding down the plane.

- (a) Draw a diagram to show the forces acting on the particle.

(1)

- (b) Find the magnitude of the normal reaction force acting on the particle.

(2)

- (c) The coefficient of friction between the particle and the plane is 0.2. Show that the magnitude of the friction force acting on the particle is 7.51 N, correct to three

significant figures.

(2)

- (d) Show that the acceleration of the particle is 4.80 m s^{-2} , correct to three significant figures.

(4)

- (e) Find the distance that the particle travels as its speed increases from 2 m s^{-1} to 10 m s^{-1} .

(3)

(Total 12 marks)

Q23.

A particle is initially at rest at the point A with position vector $(9\mathbf{i} + 10\mathbf{j})$ metres, with respect to an origin O . It moves with constant acceleration, and 10 seconds later is at the point B with position vector $(19\mathbf{i} - 25\mathbf{j})$ metres. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Show that the acceleration of the particle is $(0.2\mathbf{i} - 0.7\mathbf{j}) \text{ m s}^{-2}$.

(4)

- (b) Find the speed of the particle at B .

(4)

- (c) Verify that the particle passes through the point with position vector $(15.4\mathbf{i} - 12.4\mathbf{j}) \text{ m}$.

(4)

- (d) Draw a diagram to show the path of the particle as it moves from A to B . You should include the unit vectors $\mathbf{i}$ and $\mathbf{j}$ on your diagram.

(2)

(Total 14 marks)

Q24.

As a lift moves upwards its motion has three stages.

Stage I It accelerates uniformly, from rest, at 0.2 m s^{-2} , for 8 seconds.

Stage II It then travels at a constant speed for 3 seconds.

Stage III It then decelerates uniformly, stopping when it has travelled a further 1 metre.

- (a) (i) Find the distance travelled by the lift in stage I.

(2)

- (ii) Find the speed of the lift at the end of stage I.

(2)

- (iii) Find the total distance travelled by the lift.

(2)

- (b) Find the total time for which the lift is moving.

(3)

- (c) The lift and its passengers have a total mass of 600 kg. The lift is supported by a single cable. Find the tension in the cable when the lift is accelerating upwards at 0.2 m s^{-2} .

(3)

(Total 12 marks)

Q25.

A particle moves in a horizontal plane. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

Initially the particle is at the origin and has velocity $(5\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$. It moves with constant acceleration. At time $t = 10$ it has position vector $(40\mathbf{i} - 15\mathbf{j})$ metres.

- (a) Show that the velocity of the particle is $(3\mathbf{i} - \mathbf{j}) \text{ m s}^{-1}$ when $t = 10$.

(4)

- (b) Find the acceleration of the particle.

(3)

- (c) The mass of the particle is 15 kg. Find the magnitude of the resultant force acting on the particle.

(4)

(Total 11 marks)

Q26.

A small stone falls vertically from rest. When the stone hits the ground it is travelling at a speed of 24.5 m s^{-1} . Model the stone as a particle and assume that no resistance forces act on the stone as it falls.

- (a) Find the time that it takes for the stone to fall to the ground.

(2)

- (b) Show that the stone falls a distance of 30.625 metres.

(3)

- (c) Find the time for which the stone has been falling when it is 5 metres above ground level.

(4)

(Total 9 marks)

Q27.

A box, of mass 20 kg, is initially at rest on a rough horizontal surface. A horizontal force of magnitude P newtons is applied to the box. The coefficient of friction between the box and the surface is 0.3.

- (a) State the magnitude of the normal reaction force acting on the box.

(1)

(b) Find the magnitude of the friction force that acts on the box if:

(i) $P = 80$;

(ii) $P = 40$.

(3)

(c) Find the value of P when the box is accelerating at 0.8 m s^{-2} .

(3)

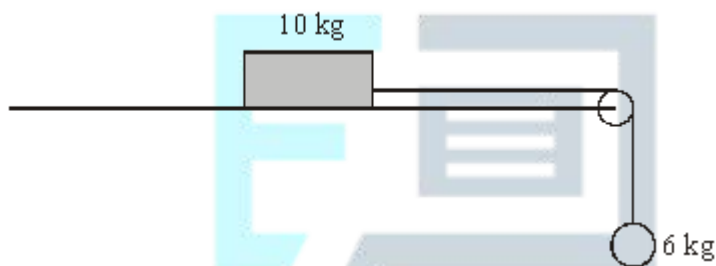
(d) When the box reaches a speed of 6 m s^{-1} , the horizontal force P is removed. Find the distance that the box travels after the force P is removed.

(5)

(Total 12 marks)

Q28.

A block, of mass 10 kg , rests on a rough horizontal surface. It is connected by a light, inextensible string to a particle of mass 6 kg . The string passes over a light, smooth pulley, so that the string hangs vertically, as shown in the diagram.



Model the block as a particle.

(a) The system is released from rest and the block travels 0.5 metres in 2 seconds. Find the acceleration of the system.

(2)

(b) Show that the tension in the string is 57.3 N .

(3)

(c) Find the coefficient of friction between the block and the plane.

(6)

(Total 11 marks)

Q29.

A particle of mass 4 kg moves on a smooth horizontal plane. It is initially at rest at the origin. A force $\mathbf{F} = (8\mathbf{i} - 12\mathbf{j}) \text{ N}$ acts on the particle for 20 seconds. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

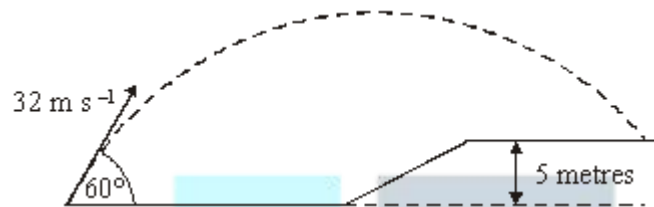
(a) (i) Find the acceleration of the particle.

(2)

- (ii) Find the velocity of the particle at the end of the 20 second period. (2)
 - (iii) Find the position of the particle at the end of the 20 second period. (3)
 - (b) At the end of the 20 second period, the force **F** is removed. Find the distance of the particle from the origin after the particle has been in motion for a total of 45 seconds. (5)
- (Total 12 marks)**

Q30.

A golf ball is hit so that it initially travels at 32 m s^{-1} at an angle of 60° above the horizontal. The ball lands on a horizontal surface 5 metres higher than the ground from which it was hit, as shown in the diagram below.



- (a) Show that the ball is in the air for approximately 5.47 seconds. (5)
 - (b) Calculate the horizontal distance travelled by the ball. (2)
 - (c) Find the speed of the ball when it hits the ground. (5)
- (Total 12 marks)**

Q31.

A train travels along a straight horizontal track. It starts from rest and accelerates at 0.5 m s^{-2} for 10 seconds.

- (a) Find the speed of the train after 10 seconds. (2)
 - (b) Find the distance that the train travels in the first 10 seconds. (2)
 - (c) The train has a mass of 200 tonnes and experiences a resistance force of 40 000 N. Find the magnitude of the forward driving force that acts on the train while it is accelerating at 0.5 m s^{-2} . (3)
- (Total 7 marks)**

Q32.

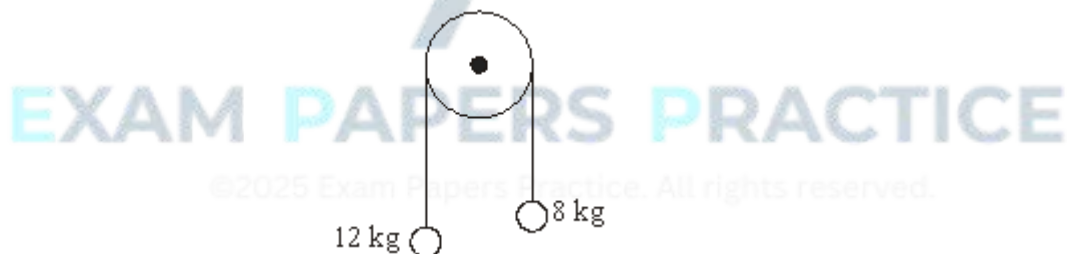
A train accelerates uniformly from rest along a straight horizontal track. After it has travelled 400 metres, its speed is 16 m s^{-1} .

- (a) (i) Show that the acceleration of the train is 0.32 m s^{-2} . (2)
- (ii) Find the time that it takes the train to travel the 400 metres. (2)
- (b) When the train has reached a speed of 16 m s^{-1} , its acceleration is increased to 0.5 m s^{-2} .
- (i) Find the distance that the train travels as its speed increases from 16 m s^{-1} to 30 m s^{-1} . (2)
- (ii) Find the total time that the train has been moving when it reaches a speed of 30 m s^{-1} . (3)

(Total 9 marks)

Q33.

A light, inextensible string has a particle of mass 8 kg attached to one end and a particle of mass 12 kg attached to the other end. The string passes over a smooth, light pulley. The particles are released from rest with the string taut and vertical on each side of the pulley. The diagram shows the pulley and the particles.



- (a) By forming an equation of motion for each particle, show that the acceleration of the particles is 1.96 m s^{-2} . (5)
- (b) Find the tension in the string. (2)
- (c) Find the time that it takes for the particles to reach a speed of 7 m s^{-1} . (2)
- (Total 9 marks)

Q34.

A model aeroplane moves in a horizontal plane with a constant acceleration. Initially the

aeroplane is at the origin and has velocity $(35\mathbf{i} + 45\mathbf{j}) \text{ m s}^{-1}$. After accelerating for 8 seconds, the velocity of the aeroplane is $(19\mathbf{i} + 13\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular and lie in the horizontal plane.

- (a) Show that the acceleration of the aeroplane is $(-2\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-2}$. (3)
 - (b) Find an expression for the position vector of the aeroplane at time t seconds. (3)
 - (c) Find the time when the position vector of the aeroplane is $(300\mathbf{i} + 225\mathbf{j}) \text{ m}$. (7)
- (Total 13 marks)**

Q35.

The graph shows how the velocity, $v \text{ m s}^{-1}$, of a cyclist varies with time, t seconds, as she moves along a straight horizontal road.



- (a) Calculate the total distance travelled by the cyclist. (3)
 - (b) Find the acceleration of the cyclist during the first 5 seconds of the motion. (2)
 - (c) As the cyclist moves, a force of magnitude P newtons acts on the cyclist in the direction of motion. A constant horizontal resistance force of magnitude 50 newtons also acts. Model the cyclist and her bicycle as a particle of mass 65 kg.
 - (i) Draw a diagram to show **all** the forces acting on the particle. (1)
 - (ii) Find the value of P during the first 5 seconds of the motion. (3)
- (Total 9 marks)**